Attention, expectation updating, and subjective models of the macroeconomy*

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Abstract

We implement an online survey to study the role of attention in macroeconomic expectation formation upon information provision. Exogenous variation in participants' subjective expectations is created by providing different professional forecasters' assessments of future inflation rate as information treatments. We find that participants paying attention to inflation respond to the information shock about inflation by significantly updating their inflation expectations, while participants paying no attention to inflation respond little. Change in inflation expectation contributes to change in macroeconomic beliefs and anticipated consumption-investment decisions. We also find evidence that respondents elicit affect-based unemployment expectation, which explains the co-movement of inflation and unemployment survey expectations. Our results demonstrate the important role of attention in households' expectation formation.

JEL classification: D12, D14, D83, D84, E31

Keywords: Expectation Formation, Rational Inattention, Consumption, Subjective model

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1 Introduction

Expectations about future economic status are of vital importance for households' decision-making because most economic decisions have an intertemporal dimension. When forming expectations, paying attention to available information plays an important role, because people need to devote attention to process relevant information in order to make precise forecasts. Paying little attention to new information leads to imprecise forecasts. For example, recent studies show that due to limited attention, professional forecasters display under-reaction to news at consensus level (see Coibion and Gorodnichenko (2012, 2015)). In this paper, we investigate the role of attention in consumers' expectation formation and their subjective models of macroeconomics.[1]

To achieve this, we use survey experiments with randomized control trials that are popularly used in recent studies.[2] In this rising literature, economists provide exogenous information to randomly assigned groups and then use this exogenous information provision to investigate decision-makers' expectation formation and economic decisions. This method helps tackle endogeneity problems in identification, such as omitted variable bias and reverse causality. Different from previous studies, in our survey experiment we further highlight whether respondents pay attention to the provided information treatments. According to rational inattention theory, if respondents pay no attention to process a certain piece of information, it will not enter the information set, and as a result, it will not affect respondents' expectation formation and economic decisions.

In the online survey, we start with a question about what information consumers pay attention to when they make the consumption-saving decision, which is one the most common and important economic decisions that people make. For those who pay attention to inflation, we provide one of two pieces of exogenous information randomly as treatment. Respondents in the "high inflation treatment" receive information of the highest inflation prediction from the Survey of Professional Forecasters (SPF) in the second

[1]The role of attention in expectation formation and decision making has been clearly discussed in the theoretical literature of rational inattention since Sims (2003), who first models information processing capacity or attention as a scarce resource. For example, Maćkowiak and Wiederholt (2015) show that consumers' inattention to the real interest rate can lead to the "hump-shaped" response of consumption to monetary policy shocks observed in the data. Gabaix (2020) shows that limited attention to future states reduces the effect of forward guidance on current economic decisions and resolves the forward guidance puzzle.

[2]See, e.g. Armantier, Nelson, Topa, van der Klaauw, and Zafar (2016), Armona, Fuster, and Zafar (2019), Coibion, Georgarakos, Gorodnichenko, and Van Rooij (2019), Coibion, Gorodnichenko, and Weber (2022), Roth and Wohlfart (2020) etc.

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quarter of 2022, whereas respondents in the "low inflation treatment" receive information from the lowest inflation prediction in SPF of that wave. Thereafter, we measure our respondents' expectations about the evolution of inflation and their personal economic situation over the next 12 months after the survey, and elicit their consumption-investment plans as well as their posterior beliefs about the personal probability of losing job, national unemployment rate and interest rate.

From data of our main wave, we find that income, liquidity asset and financial literacy are important in determining whether respondents pay attention to inflation or not. Respondents with high income are less likely to pay attention to inflation news than those with low income. However, respondents with high liquid asset are more likely to pay attention information about inflation. Second, we investigate whether paying attention to inflation matters for expectation formation as predicted by Bayesian updating. We show that respondents who pay attention to inflation information adjust their expectations partially toward the new signal (the professional forecast) but people who do not pay attention to inflation information do not adjust their expectation significantly. In order to investigate causal effects of inflation expectation update on other macroeconomic expectations and households' consumption-investment behaviors, we use the exogenous information provision as an instrumental variable for inflation expectation update. Our results show that, for respondents who pay attention to inflation information, higher inflation expectation leads to higher expectation on interest rate, more likely to purchase durable goods and non-durable goods, less likely to invest in risky asset, but also higher expectation on national unemployment and personal job-losing probability. For those who do not pay attention, we do not find such effects.

To study the persistence of treatment effects, we also conduct a follow-up survey 3 months after the first wave. In the follow-up survey, in addition to asking respondents about their macroeconomic expectations, we especially ask their actual consumption behavior in past three month.[3] We find that inflation information treatments have similar effects on macroeconomic expectations about the unemployment rate, the interest rate, and consumption-saving decisions, but with smaller size. Moreover, in the follow-up survey we also ask respondents whether inflation is important for their consumption decisions, and collect data about their knowledge regarding inflation. In line with the theoretical literature of rational inattention (see, e.g. Wiederholt (2010)), we find that the likelihood

[3]For durable goods, we ask whether they purchased durable goods in past three months. For non-durable goods or services, we asked whether they indeed consumed more of such goods and services in past three month than a period of three months before.

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of paying attention to inflation information is increasing with the importance of inflation, literacy of inflation and the subjective uncertainty in inflation.

Finally we ask why unemployment and inflation expectations comove even for individuals who think the main driving force of inflation is high demand. This contradicts the implications of short-run Phillips curve. This implies that even though people pay attention to macroeconomic shocks, they may understand propagation mechanisms of the shocks differently from classic economic theories. For example, people may think shocks that increase the inflation rate is important in their intertemporal consumption choice, but they do not thoroughly consider consequences of these shocks. Andre, Pizzinelli, Roth, and Wohlfart (2022) show that contextual cues play an important role in shaping individuals' subjective macroeconomic models. We follow this work and conduct another online survey to investigate the importance of cues or instructions in the relationship between unemployment and inflation relationships.

In our second survey, we first ask respondents their prior belief about the inflation rate in 12 months. Then we focus on two shocks on inflation: a positive demand shock (consider a situation that the inflation rate will be 10 percentage points higher than your prediction due to higher demand) and a negative oil price shock (consider a situation that the inflation rate will be 10 percentage points higher than your prediction due to oil supply disruption). We first randomly assign a shock to respondents. Next, before they make further expectations on nationally unemployment, we randomly give people in each group (demand shock group v.s. supply shock group) one of the following instructions: "think about consequence of this shock when making predictions", "use feelings and emotional reactions when making predictions", and no specific instruction ("make prediction in the usual way"). In the end we ask respondents to report their expectations about the unemployment rate.

Using data of our second survey, we show that an increase in inflation rate leads to higher unemployment expectations no matter the increase is demand-side or supply-side driven. However, the positive effects are still much larger in the case of supply-driven inflation than that of demand-driven inflation. Although people do not have a Phillips curve in mind, they still understand that higher inflation driven by higher oil price, on average, lead to more unemployment. In this part, we also investigate the role of instruction or cue in respondents' subjective models. We find that if asking respondents to use feeling or emotions to make predictions, their subjective relationship between unemployment expectations and inflation expectations are not significantly different from

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those who receive no particular instruction when making unemployment predictions. If we ask respondents to consider the consequence of a higher inflation rate when making unemployment expectations, this instruction has heterogeneous effects for supply-side-driven inflation and demand-side-driven inflation cases.

Our paper contributes to the literature that studies the role of attention in people's expectation formation and economic behaviors by using exogenous information provision. A series of papers have used information experiments to study households' beliefs about the autocorrelation of macroeconomic variables, such as Armantier, Nelson, Topa, van der Klaauw, and Zafar (2016) and Cavallo, Cruces, and Perez-Truglia (2017) for inflation expectations of consumers and Coibion, Georgarakos, Gorodnichenko, and Van Rooij (2019) for inflation expectations of firms, or Armona, Fuster, and Zafar (2019) and Fuster, Perez-Truglia, Wiederholt, and Zafar (2022) for house price expectations. Other information experiments have studied how respondents update their expectations about one macroeconomic variable in response to information about a different macroeconomic variable (e.g. Coibion, Georgarakos, Gorodnichenko, and Van Rooij (2019), Coibion, Gorodnichenko, Knotek II, and Schoenle (2020); Roth and Wohlfart (2020)). While the randomized provision of information in these experiments allows for causal identification, it misses one important aspect that whether respondents pay attention to the information provision in their everyday decision-making. In contrast to previous literature, our study starts with a direct measure of attention. This allows us to investigate the heterogeneity in people's expectation formation from the perspective of paying attention or not.

The results of this paper contributes to recent studies that use data of Michigan Survey of Consumers (MSC) to study relationships among attention, expectations and consumption attitudes. For example Binder (2018) uses news heard about interest rates to explain consumers decision-making. Pfajfar and Santoro (2013) use MSC news regarding prices to study the connection of news and expectations. Dräger and Lamla (2017) use MSC news heard data to measure attention to inflation in order to study consumers' inflation belief updating behavior. Kontny and Yin (2022) focus on specific news heard on specific macroeconomic expectation (e.g. effects of paying attention to news about price changes on inflation expectation). They also explore to what degree expectations matter for the pass-through of news attention to consumption attitudes towards durable goods, cars, and homes. One common shortcoming of all these works is that they can only study the correlation but not causal effects of one macroeconomic expectation on other macroeconomic expectations and economic behaviors. By using exogenous information provision,

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we are able to investigate the **causal relationship** between inflation expectation and other macroeconomic expectations as well as consumption-saving behaviors.

Our paper also adds to the literature that studies relationship between inflation expectation and consumption decisions. Recent studies provide mixed results concerning the relationship between inflation expectations and consumption, using qualitative measures of readiness to spend in durable goods. For example, using cross-sectional survey data of US consumers, Bachmann, Berg, and Sims (2015) observe either no relationship or, when the ZLB is binding, a negative relationship between household inflation expectations and readiness to spend on durable goods. Coibion, Georgarakos, Gorodnichenko, and Van Rooij (2019) observe sharply negative effects of inflation expectations on durable goods spending among Dutch households and modest negative effects on overall spending when inflation expectations are manipulated experimentally within a consumer survey. In contrast, Duca-Radu, Kenny, and Reuter (2021) use cross-sectional survey data from the euro area to document that higher expected inflation boosts readiness to spend in a ZLB environment. D'Acunto, Hoang, Paloviita, and Weber (2019) find that among Finnish men, only high-IQ individuals act in a manner that indicates a positive relationship between inflation expectations and the readiness to purchase durables. D'Acunto, Hoang, and Weber (2016, 2018) use data of German and Polish consumers respectively and find that an unexpected announcement of an increase in the consumption tax boosts inflation expectations and readiness to spend on durables by economically significant amounts. Burke and Ozdagli (2014) use a panel dataset and find durables spending increases with inflation expectations only for certain types of households, while non-durables spending does not respond to inflation expectations. One potential reason they give is that expectations on inflation and unemployment may move in the same direction because they also find that spending decreases with an expected increase in unemployment.

As Carlstrom and Fuerst (2008) point out, especially mark-up shocks might be problematic, as they could lead to effects on output and inflation that are not consistent with the short-run Phillips curve correlations. Consequently, Dräger, Lamla, and Pfajfar (2016) use MSC data and check for the role of supply and demand shocks. They show that consistency regarding the Phillips curve drops around 1988-1989, around 1992-1993 and around 2001 when supply shocks dominated demand shocks. Kamdar (2019) uses a rational inattention model to show that the relationship between inflation expectation and unemployment expectation may different in the case that inflation is increased due to a

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positive demand shock from that inflation is increased due to a negative supply shock. Different from the previous studies, since we focus on individuals' subjective relationship among macroeconomic expectations, we also use a measure of their perceived driving force of inflation and find comovement of unemployment and inflation expectations although the unemployment expectation is lower for those who think inflation is mainly driven by demand. In order to further investigate the comovement of unemployment and inflation expectations we conduct survey and study the role of affect in forming expectations. Therefore, our paper also contributes to the literature that studies relationship between inflation expectation and unemployment expectation under different types of inflation increase.

The rest of the paper is structured as follows: in Section 2 we describe the design of the main experiment. In Section 3, we provide details on the data collection. In Section 4, we present main results regarding attention allocation, macroeconomic expectation update. Section 5 presents the results on the causal effect of expectations about inflation on people's unemployment and interest rate expectations, personal unemployment probability and consumption-investment behaviors. Section 6 further investigates the role of inflation driving forces and instructions in the relationship between inflation expectation and unemployment expectation. Section 7 concludes.

2 Survey design

2.1 Main survey: attention, expectations, and economic behaviors

We have conducted the first wave of experimental survey on Amazon MTurk.[4] Figure A.1 in Appendix plots the flow our survey.

Attention choice: The survey starts by inviting respondents to imagine a hypothetical scenario where they *unexpectedly* receive $5,000 additional incomes today. Then they are asked to solicit what macroeconomic information they would like to acquire, when they are making consumption-saving decision with the additional incomes. A randomized list of choices is provided, including "inflation or deflation", "unemployment rate", "interest rate", "GDP growth rate" and "others". Respondents are allowed to choose multiple

[4]Recruiting online panels via Amazon MTurk is commonly used research practice in economics and other social sciences and results in more nationally representative samples than result from typical convenience sampling (Berinsky, Huber, and Lenz (2012), Casler, Bickel, and Hackett (2013), and Levay, Freese, and Druckman (2016)).

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choices.[5]

Prior beliefs: inflation expectation. Afterwards, the first assignment of theme begins. Respondents are assigned into one of two groups ("pay attention to inflation or deflation" and "pay no attention to inflation or deflation" groups) based on their previous elicitation: if a respondent only chose "inflation or deflation", they will be assigned into the "inflation or deflation" group; otherwise, they will be assigned into the other group. Next, respondents are asked to predict the rate of inflation or deflation in the next 12 months and solicit the level of uncertainty about their prediction. To further understand the mechanism on top of respondents' mind when making the pre-treatment inflation prediction, *we provided a randomized list of potential mechanisms that consists of three demand-side channels and three supply-side channels for inflation.* Respondents are required to allocate 100 points among these different channels in a way that is proportional to the importance in driving their inflation expectation.

Information treatment: professional forecasts. After the pre-treatment elicitation, respondents are randomly assigned into either a high- or a low-treatment group. The high-inflation-treatment group receives the highest inflation forecast from the Survey of Professional Forecasters (9% in 12 months), whereas the low-inflation-treatment group receives the lowest inflation forecast (0.8% in 12 months).

> "Previously you expected the inflation rate in 12 months to be **x%** in the U.S. According to a professional financial services provider that regularly takes part in a survey of professional forecasters by the Federal Reserve of Philadelphia, the inflation rate in 12 months from now will be **9 percent** in the U.S."

Macroeconomic expectations and economic behaviors. After information treatment, respondents are asked to elicit a series of posterior predictions about rates of inflation, unemployment and interest. Respondents are also asked to re-elicit the level of uncertainty about their predict of the treatment variable. Attention checkers are designed to screen out inattentive participants.[6] Afterwards, we use a series of structured questions to elicit respondents' expectation on their future financial condition, intention to buy durable goods and spending on consumption. In addition, we follow the survey

[5]One may argue that people choose to pay attention to inflation because inflation is provided in the survey. To mitigate the experimenters' demand effect, the choices are randomized.

[6]Respondents who fail the attention checkers are immediately directed to the end of the questionnaire. We also use Qualtric's bot checker to filter out bots. About 5% of respondents who entered the survey are screened out.

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design in MSC and ask their expected probability of being unemployed during the next 12 months and expected probability of finding another job that they are willing to take within three months if being unemployed. Finally, we collect a standard set of demographic (personal and household) characteristics, plus three questions about finance and economics to test their sophistication on financial markets.

Experimenter demand effect. Numerous experimental survey studies, such as those conducted by Stantcheva (2023), Andre, Pizzinelli, Roth, and Wohlfart (2022), and De Quidt, Haushofer, and Roth (2018), intentionally structure their surveys to minimize the influence of experimenter demand effects (EDE) on the survey results. We take EDE seriously. First, we implement these strategies at the survey design stage. Appendix 8.2 offers an extensive examination of the steps we have taken and the strategies we have employed to counteract the impact of EDE within our survey's design. Second, we conducted a follow-up survey and re-elicit respondents' expectation and economic behavior. Having a follow-up survey can mitigate EDE, as it is short-lived. In the following section, we explain the follow-up survey.

2.2 Follow-up survey

To examine the persistency of the effect of information treatments, three months after the main-wave survey has been completed, respondents who are assigned into the "inflation or deflation" theme are re-invited to participate in the follow-up survey. Before follow-up questions, we collect information about the **importance** of inflation in respondents' economic decision-making and respondents' **literacy of inflation** to further investigate their attention allocation behavior. Then, we ask similar questions to the main wave to elicit respondents' predictions about rates of unemployment, interest and inflation in the next 12 months. Further, they are asked whether they spent more, about the same, or less on food and leisure activities and durable goods, respectively, over the past three months. Finally, we ask their investment decisions (splitting $1,000) on stock market index funds and savings account.

3 Data

The main-wave survey was open from 31st August and closed on 1st September 2022. We recruit a sample of general US population who are representative in terms of gender, age,

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region and total household income through MTurk which is widely used in economics research. We invite not only employed people but also unemployed and self-employed to answer the survey. We dropped observations who failed attention-checker questions and answer expectation questions less than three seconds. In the end, we have 822 valid observations.

Data description. Let us briefly describe the socioeconomic and demographic characteristics, attention allocation behavior and heterogeneous expectation formation behavior. In the survey, we listed four macroeconomic variables that people may be interested in when making consumption-saving decisions. In addition to inflation and unemployment, they can also choose GDP growth rate, interest rate or any other information. On average, respondents care more about interest rate and inflation when making consumption-saving decisions, whereas information regarding unemployment and GDP growth rate attract less attention. In addition, we also observe that the average inflation expectations of our respondents are higher than the largest inflation expectation provided by the professional forecasters. However, there is a large dispersion in beliefs about the inflation rate in 12 months, ranging from 5% deflation rate to 25% inflation rate. The dispersion is much smaller for professional forecasters: ranging from 0.8% to 9% for the inflation rate. Finally, respondents, on average, think that it is a good time to spend more money on durable goods or non-durable goods (food and services), but we do not find significant difference between investing in stock-index fund or put money into saving account.

	(1)				
	No. of Observations	Mean	Standard Deviation	Minimum	Maximum
Attention_inflation	822	0.79	0.41	0.00	1.00
Attention_unemployment	822	0.22	0.41	0.00	1.00
Attention_interest_Rate	822	0.71	0.45	0.00	1.00
Attention_GDP_growth	822	0.49	0.50	0.00	1.00
Prior: inflation	822	11.72	6.47	-4.80	24.70
Financial literacy	822	2.03	0.93	0.00	3.00
Durable goods	822	0.41	0.74	-1.00	1.00
Non-durable goods	822	0.30	0.69	-1.00	1.00
Invest funds	822	499.66	214.00	0.00	1000.00

Table 1: Summary statistics.

Sample of respondents. Table 2 provides a comparison between our survey data and data from Michigan Survey of Consumers (MSC), which is a monthly-conducted survey by the university of Michigan. MSC is popularly used in recent macroeconomic literature

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of expectation formation, attention allocation and consumption decisions. In each month, about 600 representative US citizens will be interviewed via telephone. In order to show the representativeness of our sample, we make a simple comparison with the MSC data that were collected also in September 2022.

In Table 2, we notice that our data match the mean of gender, marital status and residence region very well. But we also see there are differences for age and college. In our sample, people on average are younger and more educated than those in MSC. We think a potential reason is due to the survey method, i.e. we use MTurk that is conducted online via smartphone or computer, whereas MSC is conducted via telephone.

	Main wave				
	No. of Observations	Mean	Standard Deviation	Minimum	Maximum
Age	822	37.48	11.59	20.00	84.00
Female	822	0.40	0.49	0.00	1.00
Married	822	0.78	0.41	0.00	1.00
college	822	0.92	0.26	0.00	1.00
West	822	0.26	0.44	0.00	1.00
North central	822	0.25	0.44	0.00	1.00
Northeast	822	0.09	0.29	0.00	1.00
South	822	0.39	0.49	0.00	1.00
Inflation expectation	822	11.72	6.47	-4.80	24.70
	Michigan survey of consumers (the August 2022 wave)				
	No. of Observations	Mean	Standard Deviation	Minimum	Maximum
Age	599	51.68	17.89	18.00	97.00
Female	601	0.39	0.49	0.00	1.00
Married	601	0.64	0.48	0.00	1.00
college	584	0.61	0.49	0.00	1.00
West	601	0.23	0.42	0.00	1.00
North central	601	0.26	0.44	0.00	1.00
Northeast	601	0.15	0.36	0.00	1.00
South	601	0.37	0.48	0.00	1.00
Inflation expectation	481	7.18	10.47	-30.00	50.00

Table 2: Summary statistics: comparing main wave and Michigan survey of consumers data.

Main variables. Next, we define the main variables that are used in our empirical analysis below. First, we measure whether respondents pay attention to economic information: "In your consumption-saving decision, to what information would you like to pay attention?" Then we define a dummy variable for paying attention to inflation if they choose to pay attention to information about inflation.

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Then we define a measure of expected inflation rate by asking respondents their forecast about the national inflation in 12 months. Respondents can directly choose a number about these expectations. We also collect data regarding personal expectations on the unemployment rate and the interest rate in 12 months. Moreover, we create an unweighted index of respondents' planned change in non-durable goods consumption from the four weeks prior to the main survey to the four weeks after the survey, using their qualitative spending plans for food and leisure activities. Then, for durable goods consumption, we follow the survey design in MSC and ask respondents whether now is a good or bad time for people to buy major durable goods such as furniture, cars, or household appliances. Finally, we create a measure of people's investment decisions based on their self-reported investment of 1000 dollars into saving account with fixed interest rate and a stock-index fund.

Randomization. As shown in Table 3, our sample is well-balanced for a set of key characteristics and pre-treatment beliefs about the inflation rate and the unemployment rate. The means do not differ significantly across treatment groups for any of these variables and we cannot reject the null hypothesis that the partial correlations of the variables with a dummy for being in the high inflation treatment are jointly zero.

Variable	Group 1	Group 2
Female	0.397	0.398
Married	0.871	0.859
Age	39.3	39.2
College	0.92	0.91
Household size	5.07	5.05
Prior inflation expectation	11.8	12.6
Number of observations	415	407

Table 3: Average value of main demographic characteristics and inflation expectation for different information treatment groups: Group 1 is high inflation treatment; Group 2 is low inflation treatment.

4 Attention and expectation formation

In this section we will discuss determinants of attention to inflation information and the role of attention in people's expectation updating.

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4.1 Attention devoted to inflation

Result 1. *Respondents with high income are less likely to pay attention to inflation information comparing to those with low income; respondents with more liquid asset are more likely to pay attention to inflation comparing to those with less liquid asset.*

Table 4 shows consumers' attention allocation behavior when they make consumption decisions. First, we observe that the likelihood of paying attention to inflation is negatively correlated with income. Consumers with high annual income (above 6000$) are about 12 percentage points less likely to pay attention to inflation when making consumption decisions comparing to the group with low annul income. One possible reason is that people with high income do not care so much about price change, and they also save more (see e.g. Dynan, Skinner, and Zeldes (2004)) such that they have more buffer stock. From the second row, we notice that people who are grouped to high asset are more likely to pay attention to inflation. This is possibly due to that inflation rate in the future determines the real return of their asset holding. The third row shows that people with more financial literacy are more likely to pay attention to inflation. In the literature of rational inattention, such as Fuster, Perez-Truglia, Wiederholt, and Zafar (2022) and Yin (2021), financial literacy or numeracy are usually used as indicators for attention/information-processing cost. The more financial literacy people have, the more information they can process.

In column 2, we add extra control variables into the baseline estimations, including age, gender, education, marital status, number of kids, and residence location dummy. As shown in Table 4, after controlling for employment status and residence, our main results do not change.[7]

4.2 Update of inflation expectations

Here we follow Roth and Wohlfart (2020) and first define the different between professional forecast and individual's prior belief as the information "shocks" of inflation:

$$shock_i^{infl.} = \begin{cases} 9 - \text{prior}_i^{infl.}, & \text{if} \quad \text{highinflation}_i = 1 \\ 0.8 - \text{prior}_i^{infl.}, & \text{if} \quad \text{highinflation}_i = 0 \end{cases} \tag{1}$$

[7] In the follow-up survey, we add more questions about inflation: importance of inflation information in consumption decision-making and literacy questions about inflation. As shown in Table A.1 in Appendix, importance and literacy are also positively correlated with the likelihood of paying attention to inflation. This is in line with the model prediction in the rational inattention literature, such as Wiederholt (2010).

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	(1) Attention to inflation	(2) Attention to inflation
Prior inflation uncertainty	0.0339**	0.0325**
	(2.15)	(1.98)
Financial literacy	0.0617***	0.0443***
	(4.25)	(2.87)
High income		-0.0802***
		(-2.71)
Liquid asset		0.0260**
		(2.51)
Demographic characteristics	×	✓
Adjusted R^2	0.0274	0.0567
N	822	822

t statistics in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 4: The table presents marginal effects from probit regressions: relationship between socioeconomic status and attention to specific economic news. Dependent variables are dummy variable indicate whether respondents pay attention to inflation. Main explanatory variables are income, liquid asset and financial literacy. We also control for age, gender, education, marital status, number of kids, and residence location dummy in the second column.

where $highinflation_i$ is an indicator taking value one for respondents who received the high inflation information treatment, and zero for individuals who received the low inflation treatment.

Then, we run regressions of the difference in people's posterior and prior expectations on inflation on the "shock" define above with the following equation using OLS:

$$update_i^{infl.} = \alpha_0 + \alpha_1 shock_i^{infl.} + \alpha_2 shock_i^{infl.} \times highinflation_i + \Phi_i X_i + \epsilon_i, \tag{2}$$

where X_i is a vector of control variables, including age, gender, education, employment status, income and asset ranks, financial literacy, and dummies of regions of residence. We also think individuals' prior expectation affects both the information shock and the update magnitude, and therefore, we control for people's prior belief.

Result 2. *The information provision regarding inflation strongly shifts expectations towards the professional forecast for those who pay attention to inflation information in their consumption-saving decisions.*

From column 1 of Table 5, we observe that individuals who experience a larger information shock update their inflation expectation more. This change in expectations is

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consistent with the rule of Bayesian updating, which predicts that respondents should adjust their expectations partially or fully towards new observed signals (the learning rate is between zero and one). The learning rate in our regression is about 0.15 for inflation that is in line with the prediction mentioned above.[8] These results are consistent with findings in previous literature, such as Roth and Wohlfart (2020).

However, what is new in our findings is that only respondents who pay attention to inflation adjust their expectations toward the professional forecasts (see column 2 and 4 in Table 5). These results are in line with predictions in the rational inattention literature. There is a vast amount of information that is, in principle, available to decision-makers (e.g., newspapers, all information available on the internet, etc), but due to limited attention it is not possible to attend to all of the information available. Therefore, decision-makers need to decide which information to attend to carefully, which information to attend to less carefully, and which information to ignore. If people do not pay attention to particular information, they do not incorporate this piece of information into their decision-making information-set, and thus do not adjust their expectations.

	Attention=1 Update_inflation	Attention=0 Update_inflation	Attention=1 Update_inflation	Attention=0 Update_inflation
Shock_inflation	0.152***	0.0472	0.208***	0.0846
	(4.25)	(0.69)	(5.10)	(1.05)
Prior inflation uncertainty	-0.338	0.308	-0.116	0.585
	(-1.54)	(0.46)	(-0.55)	(0.80)
Prior uncertainty*Shock			0.0579***	0.0312
			(3.22)	(0.72)
Demographic characteristics	×	×	✓	✓
Adjusted R^2	0.241	0.228	0.253	0.229
N	648	174	648	174

t statistics in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 5: The table presents results from OLS regressions: relationship between socioeconomic status and inflation expectation updating. Dependent variable is inflation expectation update (posterior inflation expectation - prior inflation expectation). Main explanatory variables are information shock (information provision-prior expectation), income, liquid asset and financial literacy. In columns 3 and 4, we also control for age, gender, education, marital status, number of kids, and residence location dummy.

Moreover, our results also have important policy implications. For example, central banks usually use inflation target to manage consumers' expectations (see e.g. Angeletos and Sastry (2021)). When policy-makers communicate, they have to keep the fact in mind

[8]Columns 3 and 4 of Table 5 checks the robustness by adding more control variables.

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that some consumers do not pay attention to the provided information and thus do not adjust their expectations. According to results in Table 4, people do not pay attention possibly due to the lack of financial literacy. Therefore, increasing consumers' financial literacy can help increase the effectiveness of policy-communication.

Following Roth and Wohlfart (2020), Coibion, Gorodnichenko, Knotek II, and Schoenle (2020), Coibion, Georgarakos, Gorodnichenko, Kenny, and Weber (2021) etc., we employ a three-month follow-up survey in which no treatment information is administered to test whether changes in inflation expectations persist. As shown in Table A.2 in Appendix, the learning rate is still positive and significant but smaller than the short-run learning rate. Moreover, learning rates still differ significantly between respondents who pay attention or pay no attention to inflation.

5 Attention and subjective macroeconomic models

Next, we will use IV regressions to investigate whether consumers think the economy as macroeconomic theories suggest, such as whether they have Phillips curve and Taylor rule in mind when making decisions. Finally, we will discuss the role of information acquisition and belief updating on consumption-investment decisions.

5.1 Empirical specification

In the previous section we presented that our respondents update their expectations on inflation rate in response to professional forecasts. This provides us with a first stage to investigate the causal effect of macroeconomic expectations on expectations about personal economic outcomes. Specifically, we study consumers' subjective economic model for respondents who pay or pay no attention to inflation. As a first step, we examine how these expectations are correlated with individual i's posterior beliefs and economic decisions about Y$\in$\{the future unemployment rate, interest rate, personal job, durable and non-durable goods consumption and investment into stock-index fund\}.

$$Y_i = \beta_0 + \beta_1 inflation_update_i + \beta_2 prior_inflation + \Pi^T X_i + \epsilon_i, \tag{3}$$

where X_i is a vector of the same variables we controlled in our previous estimations. However, the OLS estimate of β_1 cannot be directly interpreted as a causal effect for the following possible reasons. For example, it is possible that people have more optimistic

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or pessimistic attitudes and thus respond differently to questions regarding the posterior expectations and economic outcomes. The estimate of β_1 could be biased towards zero because of measurement error in the posterior belief. To deal with omitted variable bias, reverse causality and measurement error, we follow Roth and Wohlfart (2020) and instrument respondents' inflation update (difference between posterior and prior beliefs) with the shock defined above. This is because the shock is determined by random assignment to different professional forecasts. As a respondent's shock is correlated with her prior belief, we also control for the prior. We estimate the following equation using two-stage least squares:

$$Y_i = \gamma_0 + \gamma_1 \widehat{inflation_update}_i + \gamma_2 prior_inflation + \Theta^T X_i + e_i, \tag{4}$$

where

$$\widehat{inflation_update}_i = \hat{r}_0 + \hat{r}_1 shock_i + \hat{r}_2 prior_inflation + \hat{\Delta}^T X_i, \tag{5}$$

and $inflation_update_i$ is the difference between posterior and prior beliefs about inflation; $shock_i$ is the difference between professional forecast and the prior belief.

5.2 Main results

In this section, we want to show whether individuals understand macroeconomic models in a consistent way to classic economic theories. For example, do people have Phillips curve in mind? Do people know central banks follow the Taylor rule to make monetary policies? Do consumers make consumption-saving decisions according to Euler equation?

Results 3. *Paying attention to inflation helps respondents understand macroeconomic models correctly in terms of relationships between inflation expectation and expectations of other macroeconomic variables and consumption-investment behaviors.*

Phillips curve. The original Phillips curve proposed as an empirical relation by Phillips (1958) and Samuelson and Solow (1960) asserts a negative correlation between wage growth, or the general inflation rate (assuming that prices grow in line with wages, adjusted for productivity growth), and the rate of unemployment. Then Keynesian models of the 1960s typically implied that policies could forever be used to achieve lower unemployment and higher output at the cost of higher inflation. This short-run trade-off between inflation rate and unemployment have been embedded in many macroeconomic models (see e.g. Calvo (1983), Gertler and Leahy (2008), Galı and Gertler (1999), Mankiw

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and Reis (2002), Afrouzi and Yang (2021)). Dräger, Lamla, and Pfajfar (2016) use MSC data and investigate whether consumers have Phillips curve in mind by showing the correlation between inflation expectations and unemployment expectations. However, their empirical estimations suffer from endogeneity problem: people with certain unobservable characteristics may tend to report higher inflation expectations and unemployment expectations. Using data from RCT, we are able to avoid this omitted variable bias problem. In addition, the negative correlation between unemployment rate and inflation rate if higher inflation is driven by a positive demand shock. That is to say, if the inflation rate increases due to higher input cost (negative supply shock), we would expect a positive correlation between inflationand unemployment rate.

In columns 1 and 2 of Panel A in Table 6, we study respondents' subjective relationship between unemployment rate expectation and their expectations on inflation. As we control for prior expectation on inflation, the interpretation of inflation update is equivalent to the effect of posterior inflation expectation on unemployment expectation. This is in line with the fact that we conduct our survey in a period of "oil crisis" due to Russia-Ukraine war. But we also want to know the role of subjective inflation driving force in the relationship between unemployment rate and inflation rate. From the significant and negative coefficient in the third row, for a given level of posterior expectation on inflation, respondents who think the inflation are mainly driven by the demand side have a smaller expectation on unemployment.[9] This is consistent with the textbook macroeconomic model on inflation and unemployment. However, for respondents who do not pay attention to inflation, we do not find such relationship.

Inflation expectation update and personal job-losing probability. In addition to the correlation between inflation expectation update and the national unemployment rate, we also investigate the role of inflation expectation update in subjective personal unemployment probability. After the inflation information treatment, we ask respondents the probability that they will become unemployed in one year. To begin with, column 3 of Panel A in Table 6 reports results for respondents who pay attention to inflation and column 4 reports that for respondents who do not pay attention to inflation. We show in column 3 that given a level of prior inflation expectation, those who have higher posterior expectation (adjust inflation upward) believe that they are more likely to become unemployed in the future.

[9] It is also reasonable if people think inflation is mainly driven by the demand-side factors, because the US government issued several stimulus polices to deal with pandemic.

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Taylor rule. Nominal interest rate generated by Taylor rule is highly correlated to that set by the US Fed. Do people understand that Taylor rule is one of important rules that central banks adjust its policy to stabilize price? From column 5 of the upper panel in Table 6, we observe that for those who pay attention to inflation have a positive correlation between inflation expectation and interest rate expectation. This implies that if respondents expect a higher inflation in the future, they also expect that the central bank will adjust its nominal rate upward to stabilize the price. However, in column 6 we again find that for respondents who do not pay attention to inflation, this correlation is not statistically significant.

Inflation expectation and consumption decisions (Euler equation). According to modern macroeconomics, consumers make optimal consumption-saving plans based on the Euler equation. One important model prediction is that when consumers expect a higher rate of inflation in the future, they would like to consume more today due to lower real return of savings, i.e. consuming today is cheaper than consuming tomorrow. In our survey, we ask respondents to report their consumption decisions for both durable goods (such as furniture, car, and home) and non-durable goods (such as foods and leisure activities). From the first four columns of Panel B we notice that inflation expectation update are positively correlated with both durable goods and non-durable goods consumption no matter whether respondents pay attention to inflation or not. This implies that consumers make consumption plans in a consistent way with model predictions of the Euler equation.

Inflation expectation and investment choice. In the survey, we ask respondents their decision of investing how much out of 1000 dollars into stock-index fund and saving how much into saving account. From the last two columns of Panel B in Table 6, we see that for respondents who pay attention to inflation information, higher posterior inflation expectation leads to lower investment in stock-index funding. This is possibly due to that investors care about the real return of their investment, and therefore, if they expect a higher inflation in the future, they expect a lower real return and thus invest less.

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	(1) Attention=1 Unemp. expectation	(2) Attention=0 Unemp. expectation	(3) Attention=1 Losing job prob.	(4) Attention=0 Losing job prob.	(5) Attention=1 Interest expectation	(6) Attention=0 Interest expectation
Update_inflation	0.819***	-1.660	1.401***	-12.09	0.132**	-0.158
	(15.31)	(-0.74)	(3.53)	(-1.04)	(2.29)	(-0.40)
Demand driven inflation	-0.174	8.483	5.676***	62.84**	-0.309	0.797
	(-1.39)	(1.26)	(3.83)	(2.44)	(-1.54)	(0.51)
Demand driven inflation*Pre inflation	-0.0172*	-0.486	-0.517***	-3.872**	0.00527	-0.0452
	(-1.88)	(-1.13)	(-4.94)	(-2.20)	(0.40)	(-0.45)
Pre_inflation	0.868***	0.699***	3.518***	2.890**	0.118***	0.109***
	(36.75)	(3.32)	(32.22)	(2.04)	(10.65)	(3.22)
Adjusted R^2	0.627	.	0.574	.	0.343	0.0556
N	648	174	647	174	648	174

	(1) Attention=1 Non-durable goods	(2) Attention=0 Non-durable goods	(3) Attention=1 Durable goods	(4) Attention=0 Durable goods	(5) Attention=1 Invest Funds	(6) Attention=0 Invest Funds
Update_inflation	0.0999***	0.114	0.0729**	-0.113	-21.21*	-16.62
	(2.61)	(0.58)	(1.98)	(-0.47)	(-1.66)	(-0.25)
Demand driven inflation	-0.106	-0.0528	-0.0858	1.194	-8.590	227.0
	(-0.81)	(-0.07)	(-0.68)	(1.29)	(-0.21)	(0.84)
Demand driven inflation*Pre inflation	-0.000177	0.0111	-0.00394	-0.0727	0.163	-12.10
	(-0.02)	(0.22)	(-0.51)	(-1.23)	(0.07)	(-0.72)
Pre_inflation	0.0545***	0.0358**	0.0547***	0.0434**	-4.143*	2.650
	(6.91)	(2.11)	(7.25)	(2.08)	(-1.65)	(0.43)
Adjusted R^2	0.174	.	0.308	.	.	0.0601
N	648	174	648	174	648	174

t statistics in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 6: The table presents results IV regressions: relationship between inflation expectations and macroeconomic expectations (unemployment rate and interest rate), personal unemployment probability expectation, as well as personal economic behavior (durable and non-durable goods consumption and stock-index investment). Main explanatory variables are inflation update, prior belief about inflation, a dummy variable indicating demand side is the main driving force of inflation, and an interaction term of prior belief and the dummy variable. In columns 1, 3, and 5, we use observations that pay attention to inflation, whereas in columns 2, 4, and 6 we focus on observations who do not pay attention to inflation. We also control for age, gender, education, marital status, number of kids, and residence location dummy. For expectation on national unemployment rate and personal job-loss probability, we cluster standard errors at employment status.

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5.3 Persistence of treatment effects

To investigate the persistent effect of information provision, we conduct a follow-up survey 3 months after the main wave. In Table A.3, we show reduced-form estimates obtained from regressing the different outcome variables on an indicator for the "high inflation" treatment and a set of controls. For respondent who pay attention to inflation information and receive a higher professional forecast about inflation rate report a significantly higher unemployment rate. Although the sign of this effect is the same as in the main study, the size of the effect is about 50 percent of the original effect size in the main study. The treatment effects on interest rate expectation, durable and non-durable goods consumption and stock-index fund investment are also very similar to those in the main survey, but these estimates are less precise and effects are smaller than those in the main study. The difference between the main study and follow-up study occurs potentially due to that the follow-up survey started 3 months later after the main survey and people may receive other relevant signals about the macroeconomic and personal situations. Another possible reason is that we have a smaller sample of respondents that were called back to complete the follow-up survey. Therefore, these results still show that the treatment effect of information provision is persistent. We can conclude that, similarly as Roth and Wohlfart (2020), the persistence of the treatment effects suggests that our information treatment leads individuals to truly update their beliefs, while concerns about numerical anchoring, short-lived emotional responses to the treatment, or experimenter demand are mitigated.

6 Investigating mechanism between inflation and unemployment expectations

From our results above, it seems that paying attention to inflation only plays a big role in the relationship between the inflation expectation and the unemployment expectation: for those who pay attention to inflation, higher inflation expectation leads to higher unemployment expectation. In addition, we also find that given prior inflation expectation, those who think that inflation is mainly driven by demand have a lower unemployment expectation. However, this still shows that that the unemployment expectation and the inflation expectation move to the same direction. Therefore, we conduct another survey after the first wave and investigate why consumers' inflation expectations and unem-

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ployment expectations move in the same direction. Particularly, we test whether this co-directional movement between survey expectations of inflation and unemployment is a deliberate evaluation of the possible consequences or an affective response?

Respondents' affect-based reactions to a specific event may condition the probabilities that they assign on possible consequences. For example, when respondents are evaluating the effect of a higher supply-side inflation (due to wars) on unemployment, their affect might dictate their evaluation: "I dislike inflation; therefore, there is a strong likelihood that a higher inflation will lead to something undesirable." In this case, affect toward the event determines the perceived consequences; thoughts about outcomes arise as rationalizations.

6.1 Survey design

In November 2022, we conducted our second survey with 1,500 representative respondents in terms of characteristics. In this wave of survey, we design a 2×3 survey experiment, in which respondents receive an inflation scenario with randomized (i) source of future inflation and (ii) instruction on how to make their forecasts. We collect respondents' unemployment expectation and other macroeconomic expectations in the next 12 months before presenting the scenario and instruction.

Similar to wave 1, we design the source of higher future inflation to be either a stronger domestic demand for goods and service in the US or disruption in oil supply chains due to the Russian-Ukraine war. All respondents receive the scenario that "inflation rate in US unexpected rises by 10 percentage points further over the next 12 months" but are randomly given different causes based on the group assignment. Respondents assigned into demand-side inflation groups are told that the inflation raises due to a higher demand for goods and services. By contrast, respondents in supply-side inflation groups receive a scenario that the higher inflation is driven by severe disruption in oil supply chains due to the escalation of the Russian-Ukraine war. Wave 1 shows that, to some extend, different causes (i.e. demand or supply side) of future inflation anticipated by respondents impact their predictions of future unemployment, under a randomized control trial, where respondents' prior or information set is not controlled. At the beginning of this wave, respondents are firstly informed of recent values and familiarized with relevant definitions, so that they can make prediction under a common ground.

The key to our design, which allows us to disentangle whether people deliberate consequences or, more simply, just react based on their affect when making predictions of

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unemployment rate given an higher inflation, is the randomized division of the sample further into three groups with different instructions. The first treatment group (henceforth labeled as A, or affect group) receives these instructions:

> "We are interested in how you would predict future economic conditions under a certain economic environment. On the next page, you will be asked to consider a scenario first and then asked to make a series of predictions for future economy. There is no right or wrong answers to the predictions.
>
> Note: We have found that people's predictions are most accurate when they do not think about complicated consequences, but, rather, simply base their predictions on the feelings or emotional reactions they experience when they read the statement. Therefore, when you are making the following predictions on the next page, try to base your responses only on your feelings, or the emotions you experience when you imagine the scenario, rather than think carefully about all of its possible consequences."

The second treatment group (labeled as C, or consequence, group) reads the identical first paragraph but a very different second one:

> "Note: We have found that people's predictions are often most accurate when they think thoroughly about the potential consequences of an event, rather than simply based on their feelings or emotions. Therefore, when making the following predictions on the next page, try to think carefully and thoroughly about what consequences it would have if the event described were to occur, and base your predictions on the desirability of these consequences."

The control group (labeled as NI, or no instruction, group) is given the same first paragraph but are told to make predictions in their usual way. The NI groups, where all respondents still receive an instruction, allows us to control the survey structure (e.g. length of instruction and number of questions) and mitigate the side effects (such as priming effect) of providing an instruction on respondents' predictions.

After presenting the scenario and instructions, we re-elicit participants' 1-year ahead unemployment expectations. Figure A.2 in Appendix illustrates the flow of the core survey design.

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6.2 Results

Results 4. *Considering the consequence of different types of inflation (demand or supply-driven) matters in terms of the relationship between the unemployment rate expectation and the inflation rate expectation.*

Our main interest is the comparison between the NI group and the two instruction groups given a same type of inflation. If respondents make predictions of the unemployment rate principally based on their affect towards the inflation of a certain type, then the average response of NI group should be significantly different from the consequence group, but not from the affect group. This opposite pattern would suggest respondents consider consequences about the higher inflation when making predictions of unemployment.

Figure 1 plots density distributions of respondents' pre-instruction expectations of unemployment and inflation (the first row) for all groups, post-instruction expectations of unemployment (the second row), and revisions to unemployment expectations (the third row) for demand- and supply-side inflation groups, respectively. It demonstrates that while the distributions of unemployment expectation (and inflation expectation) before presenting the scenario and instruction are similar across all groups, post-instruction unemployment expectation and the revision vary quite significantly. Looking at the revision to unemployment expectation, distributions of the revision are widely dispersed even for respondents who are given an identical inflation scenario and instruction. This shows the existence of large heterogeneity in interpreting the scenario across respondents. More interestingly, the distributions of the unemployment expectation revision of NI groups largely resemble those from A groups not C groups, under both demand-side and supply-side inflation scenarios. Apparently the survey participants elicit their affect-based responses of expected unemployment rate, when are presented with a scenario with a higher inflation, even though respondents make heterogeneous affect-based responses under both scenarios with different types of inflation.

Table A.4 in Appendix reports statistical tests results, to examine the similarity in averages and distributions of expected unemployment revision (or post-instruction expected unemployment), respectively, across different groups. Statistical test results yield the same conclusion as we expected. First, ANOVA tests show that the difference in average revision to unemployment expectation is not significant between A and NI groups, while it is significant between C and NI groups. In addition, Kolmogorov-Smirnov test shows that the distribution of revision to unemployment expectation of NI group largely

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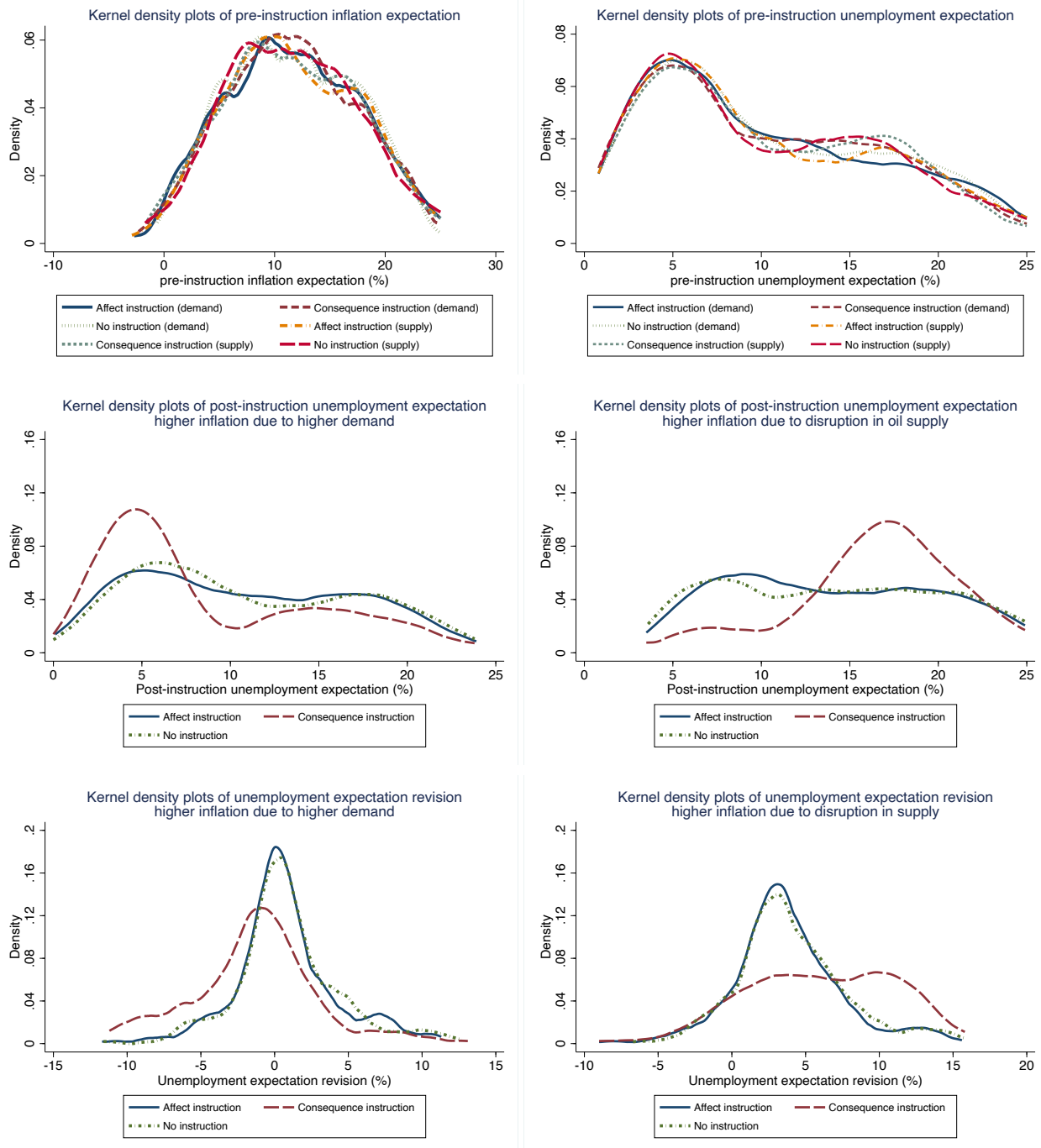

Figure 1: Unemployment expectations and inflation scenarios

assembles the corresponding distribution of A group but not C group. The evidence suggests that respondents elicit their unemployment expectations largely based on their affect towards the inflation.

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We also regress unemployment expectation revision on the group assignment, separately for groups presented with demand- and supply-side inflation scenarios. Outcomes are displayed in Table 7, where the corresponding NI group is used as the base group and all other coefficients reported are relative to the base. First, the average unemployment expectation revision of affect groups is not statistically different from that of corresponding NI groups, while the difference is significant between C and NI groups. This is consistent with ANOVA test results that respondents, without specific instruction, make affect-based unemployment expectation given a scenario regarding inflation. Second, respondents in the C group predict significantly lower rate of unemployment than the other two groups, given a higher inflation driven by demand side, whereas predict significantly higher rate of unemployment, given a higher inflation driven by supply side.

Dependent variable:	Unemployment expectation revision	
	Demand-side inflation	Supply-side inflation
	(1)	(2)
Affect (Group 1)	-0.378	
	(-1.20)	
Consequence (Group 2)	-2.584***	
	(-7.55)	
Affect (Group 4)		-0.291
		(-0.94)
Consequence (Group 5)		1.947***
		(5.91)
NI group	2.297***	5.248***
	(2.59)	(6.00)
Adjusted R^2	0.318	0.360
N	739	767

t statistics in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 7: The table presents results from OLS regressions: relationship between inflation and unemployment expectations. In column 1 we provide respondents a scenario that inflation increases by 10% due to higher demand. In column 2 we provide respondents a scenario that inflation rate increases by 10% due to negative supply shock (higher oil price, interrupted supply chain).In each specification, control for age, gender, education, marital status, number of kids, and residence location dummy. For expectation on national unemployment rate and personal job-loss probability, we cluster standard errors at employment status.

The preceding analysis leaves one question unanswered: Did the instruction manipulation on how to make prediction in fact work? To address this question, we have two designs. The first design is that, after the instruction, respondents are all asked to complete an attention checker regarding how they are asked to make predictions in the follow-

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Dependent variable:	Time spent on predictions	
	Demand-side inflation	Supply-side inflation
	(1)	(2)
Affect (Group 1)	-0.100	
	(-0.03)	
Consequence (Group 2)	24.725***	
	(9.88)	
Affect (Group 4)		2.210
		(2.498)
Consequence (Group 5)		25.936***
		(10.30)
NI group	26.196***	22.689***
	(4.21)	(2.71)
Adjusted R^2	0.404	0.367
N	739	767

t statistics in parentheses. ,
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 8: The table presents results from OLS regressions: relationship between inflation and time spent on making post-instruction predictions. In column 1 we provide respondents a scenario that inflation increases by 10% due to higher demand. In column 2 we provide respondents a scenario that inflation rate increases by 10% due to negative supply shock (higher oil price, interrupted supply chain).In each specification, control for age, gender, education, marital status, number of kids, and residence location dummy. For expectation on national unemployment rate and personal job-loss probability, we cluster standard errors at employment status. Time (seconds) spent on predictions are winsorized at top 5% and bottom 5%

ing survey questionnaire. This screens out inattentive participants and reduces noise in responses. Second, we also record the time spent on making predictions for each respondent when they were completing the relevant section of the questionnaire. Since deliberating the consequences of inflation should require more time than reflexive affect-based reaction, respondents should spend more time when making predictions. To mitigate the effect driven by extreme outliers, the average response time is winsorized at bottom and top 5%. The corresponding NI group is used as the base group and all other coefficients reported are relative to the base. Table 8 reports the average response time (in seconds) of each group, respectively for both inflation scenarios. Not only the average response time of C group is around twice as much as corresponding A group, but also the latter falls much closer to the corresponding NI group. This also indicates the similar patterns of prediction-making between A and NI groups.

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7 Conclusion

This paper aims at investigating how paying attention and cue matter in people's subjective models, i.e. relationships among macroeconomic expectations. We collect data from two information experiments. In the first survey, we first ask to what information do respondents pay attention when making consumption-saving decisions and then for those who pay attention to inflation we provide respondents with different professional forecaster's predictions about the inflation rate in 12 months. To study the importance of attention allocation in expectation formation and economic decisions, we also provide the same treatment for a group of respondents who do not pay attention to inflation. Then, we further want to test whether providing extra cue in the survey matters for the consistency of subjective model with economic theories. Therefore, in the second survey, we designed two scenarios of an increase in inflation driven by a positive demand shock and a negative supply shock respectively. We ask some respondents to make predictions on the national unemployment with emotions and some others to make predictions with considering the consequences of higher inflation rate and the rest with no instruction.

From data of our first survey, we find that income, liquid asset as well as financial literacy matter for paying attention to inflation and unemployment. Then we find that paying attention to inflation or not is important for people's expectation formation. For example, respondents who pay attention to inflation update their expectation regarding inflation after observing the new signal (professional forecaster's prediction). However, we do not find the same results for those who do not pay attention to inflation. In addition, results from IV regressions show that respondents who pay attention to inflation have more consistent subjective model with economic theories. From data of our second survey, we first show that respondents, to some extent, can distinguish effects of demand-pull inflation and cost-push inflation on unemployment. On average, the positive correlation between inflation expectation and unemployment expectation is much stronger in the case higher inflation due to higher oil price than in the case higher inflation due to higher demand. Moreover, we also show that providing respondents with different cues or instructions also matters for the relationship mentioned above. It turns out that if asking individuals to make prediction based on feeling and emotions, their subjective relationship between inflation and unemployment does not differ from that of individuals who got no instruction. However, respondents who were asked to make predictions with considering consequences of a higher inflation have more consistent subjective Phillips curve than those who got instruction that makes predictions based on emotion and no instruction.

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8 Appendix

8.1 Survey flows

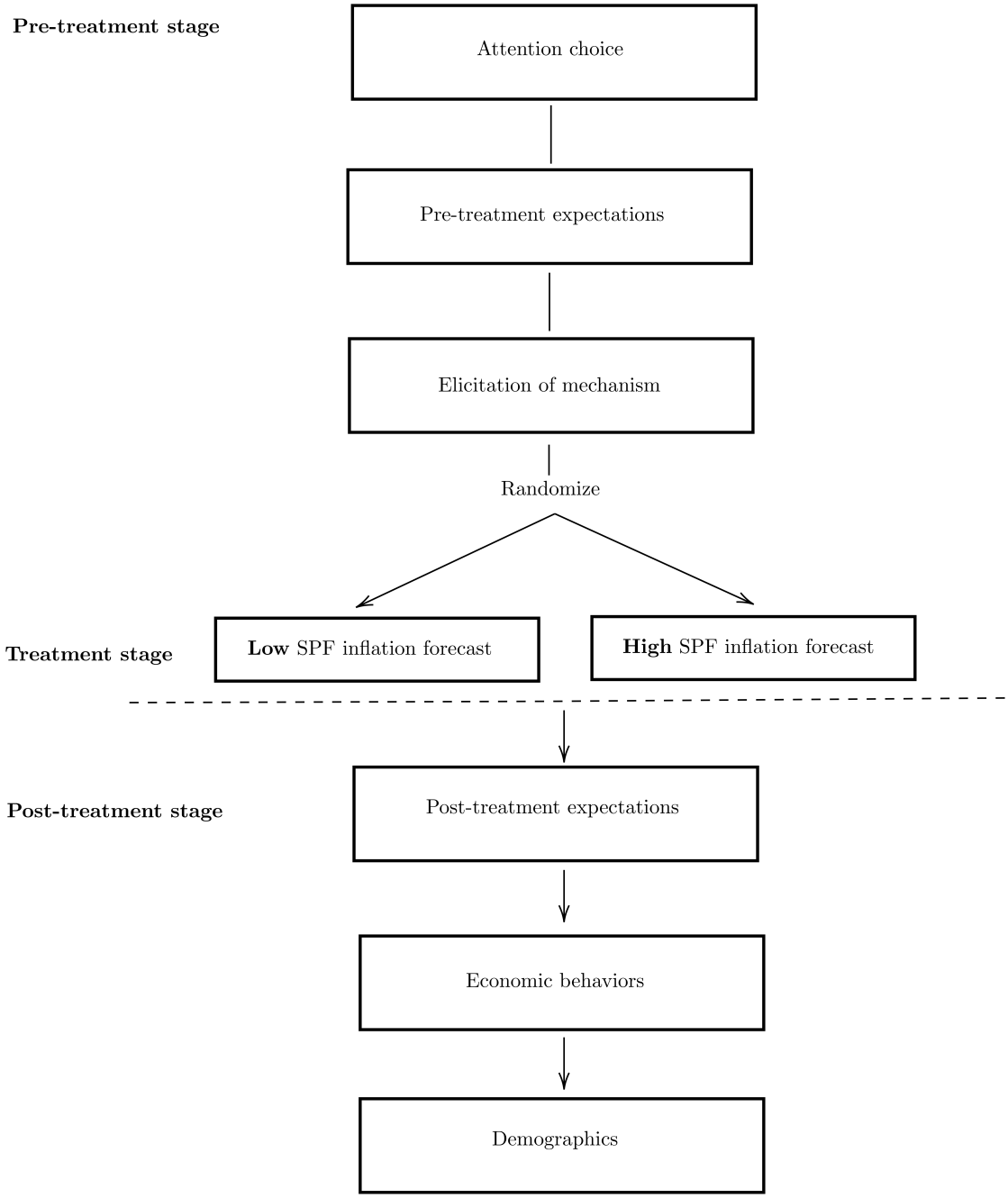

Figure A.1: Wave 1 survey design flow

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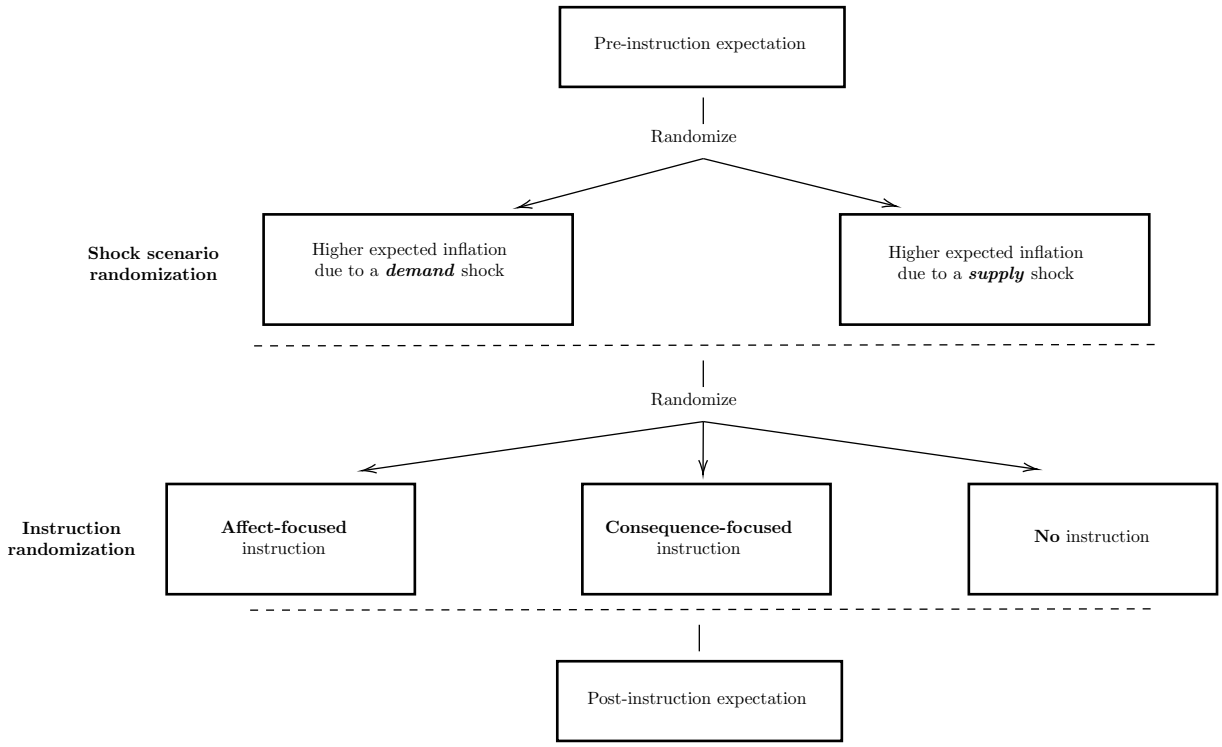

Figure A.2: Wave 2 survey design flow

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8.2 Experimenter Demand Effects

We detail our strategies and actions to counteract experimenter demand effects (EDE) within our survey design. Firstly, we mitigate EDE by collecting anonymous responses, allowing consumers to participate from their preferred location in complete privacy, free from social pressure. Secondly, our survey questions are neutrally framed, employing symmetric formulations for mechanism and evaluation queries to ensure no inherent bias towards any particular direction (e.g., "Do you think you will be financially better off or worse off...?" and "Do you think now is a good or bad time...?"). We also emphasize to respondents that questions related to their prior and posterior expectations have no definitive right or wrong answers. Thirdly, our survey design intentionally obscures the information provision, making it unclear what the actual primary "treatment" is. We offer explanations and information to households at various points before the treatment, such as clarifying the definition of inflation or deflation. Additionally, having multiple questions on the same decision screen subtly alters the contextual cue exposure, further mitigating EDE concerns.

8.3 Additional results

	(1) Attention to inflation	(2) Attention to inflation
Importance of inflation	0.0534***	0.0535***
	(12.85)	(11.82)
Literacy of inflation	0.0605***	0.0617***
	(4.63)	(4.48)
Subjective uncertainty in inflation	0.0263*	0.0276*
	(1.90)	(1.91)
Extra controls	×	✓
Pseudo R^2	0.685	0.702
N	327	327

t statistics in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.1: The table presents marginal effects from probit regressions: determinants of paying attention to inflation. Dependent variables are dummy variable indicate whether respondents pay attention to inflation. Main explanatory variables are subjective importance of inflation in decision-making, literacy regarding inflation and prior uncertainty regarding inflation. For each specification we control for income, liquid asset, age, gender, marital status and education. We also control for employment status, number of kids and residence location dummy in Column 2 as a robustness check.

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	(1) Update_inflation	(2) Pay attention=1	(3) Pay attention=0
Shock_inflation	0.120*	0.122*	0.0747
	(1.80)	(1.76)	(1.22)
High income	1.634***	1.290*	0.0858
	(2.60)	(1.92)	(0.14)
Liquid asset	-0.449**	-0.293	0.0702
	(-2.48)	(-1.51)	(0.44)
Adjusted R^2	0.567	0.530	0.972
N	327	254	73

t statistics in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.2: The table presents results from OLS regressions: relationship between socioeconomic status and inflation expectation updating. Dependent variable is inflation expectation update (posterior inflation expectation - prior inflation expectation). Main explanatory variables are information shock (information provision-prior expectation), income, liquid asset and financial literacy. We also control for age, gender, education, marital status, number of kids, and residence location dummy.

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	Unemp. expectation		Interest expectation		Dur		Non-dur		Investment	
Main wave	Attention=1	Attention=0	Attention=1	Attention=0	Attention=1	Attention=0	Attention=1	Attention=0	Attention=1	Attention=0
Inflation: High	1.044*	-0.178	0.172*	-0.0233	0.100*	-0.0511	0.132**	0.0548	-37.23*	-26.99
	(3.12)	(-0.16)	(1.90)	(-0.15)	(1.80)	(-0.53)	(2.42)	(0.67)	(-1.90)	(-0.76)
Adjusted R^2	0.661	0.725	0.361	0.546	0.333	0.239	0.319	0.254	0.0640	0.115
N	510	149	510	149	510	149	510	149	510	149

t statistics in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

	Unemp. expectation		Interest expectation		Dur		Non-dur		Investment	
Follow-up wave	Attention=1	Attention=0	Attention=1	Attention=0	Attention=1	Attention=0	Attention=1	Attention=0	Attention=1	Attention=0
Inflation: High	0.529*	0.639	0.141*	0.0655	0.146	-0.00136	0.149*	-0.0162	-28.24*	-65.54**
	(3.57)	(2.18)	(1.83)	(1.29)	(1.41)	(-0.01)	(1.68)	(-0.09)	(-1.88)	(-2.27)
Adjusted R^2	0.185	0.590	0.511	0.935	0.150	0.196	0.222	0.316	0.233	0.691
N	254	73	254	73	254	73	254	73	254	73

t statistics in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.3: Persistence of treatment effects. Dependent variables are macroeconomic expectations (unemployment rate and interest rate) and personal economic behavior (durable and non-durable goods consumption and stock-index investment). Main explanatory variable is high inflation treatment, a dummy variable indicating whether respondents obtained an exogenous treatment of high inflation expectation from professional forecasters. In columns 1, 3, 5, 7, and 9 we use observations that pay attention to inflation, whereas in columns 2, 4, 6,8 and 10 we focus on observations who do not pay attention to inflation. We also control for financial literacy, age, gender, education, number of kids, and residence location dummy. For expectation on national unemployment rate and personal job-loss probability, we cluster standard errors at employment status.

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	Subsample		
	Higher expected inflation due to ***higher demand***		
	Affect	Consequence	No instruction
Expected unemployment revision	0.66	-1.61	0.92
P value of			
ANOVA test for mean	$\overline{X}_{NI} = \overline{X}_A$, p = 0.44; $\overline{X}_{NI} = \overline{X}_C$, p = 0.00		
Kolmogorov-Smirnov test for distribution	$f_{NI} = f_A$, p = 0.94; $f_{NI} = f_C$, p = 0.00		
Post-instruction expected unemployment	11.02	8.74	11.29
P value of			
ANOVA test for mean	$\overline{X}_{NI} = \overline{X}_A$, p = 0.63; $\overline{X}_{NI} = \overline{X}_C$, p = 0.00		
Kolmogorov-Smirnov test for distribution	$f_{NI} = f_A$, p = 0.54; $f_{NI} = f_C$, p = 0.00		
n =	256	247	236
	Higher expected inflation due to ***disruption of oil supply***		
	Affect	Consequence	No instruction
Expected unemployment revision	3.92	6.12	4.27
P value of			
ANOVA test for mean	$\overline{X}_{NI} = \overline{X}_A$, p = 0.31; $\overline{X}_{NI} = \overline{X}_C$, p = 0.00		
Kolmogorov-Smirnov test for distribution	$f_{NI} = f_A$, p = 0.85; $f_{NI} = f_C$, p = 0.00		
Post-instruction expected unemployment	14.25	16.44	14.64
P value of			
ANOVA test for mean	$\overline{X}_{NI} = \overline{X}_A$, p = 0.47; $\overline{X}_{NI} = \overline{X}_C$, p = 0.00		
Kolmogorov-Smirnov test for distribution	$f_{NI} = f_A$, p = 0.59; $f_{NI} = f_C$, p = 0.00		
n =	257	241	260

Table A.4: ANOVA test for mean and two-sample Kolmogorov-Smirnov test for equality of distribution

8.4 Wave 1 main survey designs

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The main part of the survey begins when you press on the next button.

We would like to ask you about your **consumption-saving decision**, and your perspective on **future economic conditions**.
These questions have no right or wrong answers, and we are interested in your views and opinions.

Please be careful in what you choose and write, as **you will need to answer questions later based on them.**

Consumption-saving decision (multiple choices question)

Q1. Imagine you have **unexpectedly** received **$5,000 additional incomes** today.

You can consume a proportion of the $5,000 today and save the rest for the future. When you are making this consumption-saving decision, what information in the following will you look at?

*Please **tick all** that apply.*

- [ ] Interest rate
- [ ] GDP growth rate
- [ ] Inflation or deflation (i.e. percentage change in the overall price level)
- [ ] Unemployment rate
- [ ] Others

Inflation forecast treatment groups - prior and prior uncertainty

Inflation

Q3A. Inflation is the percentage rise in prices overall in the economy. Deflation (the opposite of inflation) corresponds to when overall prices are falling.

What do you think the **inflation or deflation rate** in the US will be over the next 12 months from now? (Slide to a positive rate if you believe inflation, negative value if deflation or 0 if no change in price level.)

in %

-5 0 5 10 15 20 25

Rate of inflation or deflation

Q3B. How ***certain*** are you about your forecast of inflation or deflation rate in the previous question?

- ◯ very certain
- ◯ somewhat certain
- ◯ somewhat uncertain
- ◯ very uncertain

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What contribute to your inflation expectation?

Q3C. Above you mentioned the inflation rate in the next 12 months will be **percent**. **How much do you think each of the factors in the list below contributed to your inflation expectation.** Please allocate 100 points between the different factors proportional to their importance in driving your inflation expectation/

Note: The percent must be a number between 0 and 100 and the sum of your answers must add up to 100.

Change in firms' production cost due to decrease/increase of labor force, e.g. some workers dropping out of/entering the labor force.	0
Change in firm's production cost due to global supply chains, e.g. global supply chains are disrupted/strengthened.	0
Change in demand for goods and services among households and investment demand among firms due to the Federal Reserve changes interest rate.	0
Change in demand for goods and services due to government policy change, e.g. government spending programs, stimulus payments, and higher/lower tax rates.	0
Change in demand for goods and services due to the business conditions, e.g. change in income, spending opportunities, optimism about the future.	0
Change in energy price, e.g. higher/lower price of oil and natural gas.	0
Total	0

On the next slide, we will provide you with information on the view of a professional forecaster on the inflation rate over the next 12 months. We would like to ask you to take a moment to review the information carefully.

Please note that this information is only shown once, and you will not be able to come back to it.

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High inflation forecast treatment

Please read below carefully!

You answered that you think the inflation rate over the next 12 months would be **percent.** We now would like to provide you with information on the view of a professional forecaster on the inflation rate over the next 12 months in US:

Information from professional forecasters

According to a **professional financial services provider** that regularly takes part in a survey of professional forecasters by the Federal Reserve of Philadelphia, **the inflation rate over the next 12 months will be 9 percent**.

[The next button will appear after 10 seconds.]

Low inflation forecast treatment

Please read below carefully!

You answered that you think the inflation rate over the next 12 months would be **percent.** We now would like to provide you with information on the view of a professional forecaster on the inflation rate over the next 12 months in US:

Information from professional forecasters

According to a **professional financial services provider** that regularly takes part in a survey of professional forecasters by the Federal Reserve of Philadelphia, **the inflation rate over the next 12 months will be 0.8 percent**.

[The next button will appear after 10 seconds.]

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Interest rate - new information

Q5. Given the new information, what do you think this **federal funds rate** in the US will be in 12 months from now?

in %

0 0.5 1 1.5 2 2.5 3 3.5 4 4.5 5

Federal funds rate

GDP growth rate - new information

Q6. In THIS question, you will be asked about the probability **(PERCENT CHANCE)** of something happening. *The percent chance must be a number* ***between 0 and 100*** *and the* ***sum of your answers must add up to 100****.*

Given the new information, what do you think is the *percent chance* that, over the next 12 months...

the rate of **GDP decline** will be 12% or more	0
the rate of **GDP decline** will be between 8% and 12%	0
the rate of **GDP decline** will be between 4% and 8%	0
the rate of **GDP decline** will be between 2% and 4%	0
the rate of **GDP decline** will be between 0% and 2%	0
the rate of **GDP growth** will be between 0% and 2%	0
the rate of **GDP growth** will be between 2% and 4%	0
the rate of **GDP growth** will be between 4% and 8%	0
the rate of **GDP growth** will be between 8% and 12%	0
the rate of **GDP growth** will be 12% or more	0
Total	0

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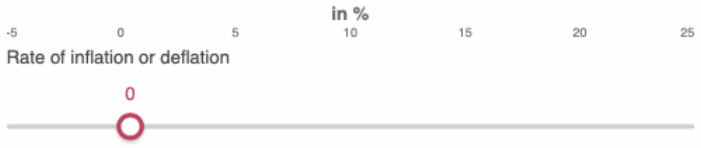
Inflation forecast treatments - posterior and posterior uncertainty

Inflation - new information

Q7A. Now given the new information, what do you think the **inflation or deflation rate** in the US will be over the next 12 months from now?

Note: previously you answered that you think the inflation rate over the next 12 months would be percent (default value of the slider below).



Q7B. How ***certain*** are you about your forecast of inflation or deflation rate?

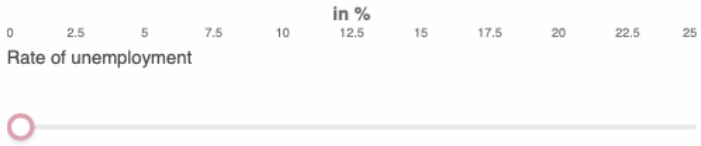
- very certain
- somewhat certain
- somewhat uncertain
- very uncertain

Unemployment forecast treatments - posterior and posterior uncertainty

Unemployment - new information

Q7A. Now, given the new information, what do you think the **rate of unemployment** in the US will be in 12 months from now?

Note: previously you answered that you think the unemployment rate in 12 months would be percent (default value of the slider below).



Q7B. How ***certain*** are you about your forecast of unemployment rate?

- very certain
- somewhat certain
- somewhat uncertain
- very uncertain

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Expectation on Economic Condition

Looking ahead, do you think you (and any family living with you) will be **financially better or worse off** over the next 12 months from now than you are these days?

- ◯ Financially better off
- ◯ Financially no difference
- ◯ Financially worse off

Generally speaking, do you think now is a **good or bad** time for people to **buy** major **durable goods** such as **furniture, cars, or household appliances**?

- ◯ Good time to buy
- ◯ Uncertain; depends
- ◯ Bad time to buy

What do you think is the **percent chance** that **you will be unemployed** during the next 12 months?

in %

0 10 20 30 40 50 60 70 80 90 100

Chance of losing main job

Suppose you were to lose your main job during the next 12 months. What do you think is the **percent chance** that **you will find another job that you can accept** (considering the pay and type of work) within **3 months** of losing your job?

in %

0 10 20 30 40 50 60 70 80 90 100

Chance of finding another job you can accept

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Compared to what you have been doing over the last 4 weeks, over the next 4 weeks do you intend to **spend more, less or about the same** on **food and leisure** activities?

- ◯ Spend more
- ◯ Spend same
- ◯ Spend less

Investment decision

Consider a situation where you have to decide how to invest **$1,000** for one year. You can choose between two possible investments. The first is a fund that **pays an annual return equal to the growth in the stock market index**. The second is a ***savings account*** that **pays 2% of interest per year**. What proportion of the $1,000 would you put into the two investments? Make sure both answers sum up to $1,000.

Stock-market-index funds	0
Savings account	0
Total	0

Financial literacy questions

G. Suppose you put $100 into a saving account and the interest rate is fixed at 3% per year for the next 5 year. After the next 5 years, how much do you think you would have in the account if you left the money to grow?

- ◯ more than $103
- ◯ exactly $103
- ◯ less than $103
- ◯ don't know

H. Imagine that the interest rate on your saving account is 3% per year and inflation is 2% per year. After 1 year, how much would you be able to buy with the money in this account, comparing with today?

- ◯ more than today
- ◯ exactly the same as today
- ◯ less than today
- ◯ don't know

I. Please indicate whether the statement is true or false: "buying a stock mutual fund usually provides safer return than buying a single company's stock"

- ◯ true
- ◯ false
- ◯ don't know

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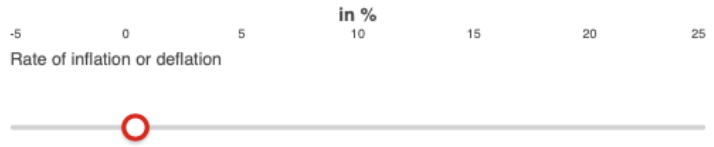
8.5 Wave 2 main survey designs

Inflation

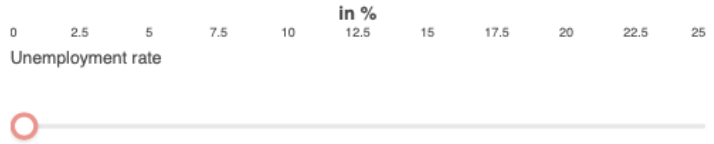
Inflation is the percentage rise in prices overall in the economy. Deflation (the opposite of inflation) corresponds to when overall prices are falling.

Q1. What do you think the **inflation or deflation rate** in the US will be over the next 12 months from now? (Slide to a positive value if you expect inflation or a negative value if deflation.)



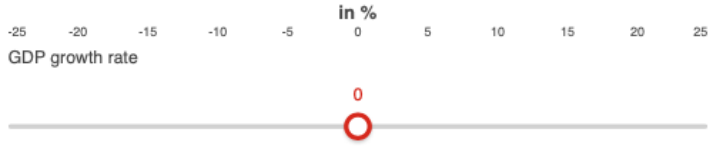
Unemployment

Q2. What do you think the **unemployment rate** in the US will be over the next 12 months from now?



GDP growth

Q3. What do you think the **growth rate of GDP** in the US will be over the next 12 months from now?



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Affect-focused instruction group

We are interested in **how you would predict future economic conditions under a certain economic environment**. On the next page, you will be asked to consider a scenario first and then asked to make a series of predictions for future economy. There is no right or wrong answers to the predictions.

Below is our instruction, **please read it carefully.**

Instruction:

We have found that people's predictions are most accurate when they do not think about complicated consequences, but, rather, simply base their predictions on the feelings or emotional reactions they experience when they read the statement. Therefore, when you are making the following predictions on the next page, **try to base your responses only on your feelings, or the emotions you experience when you imagine the scenario**, rather than think carefully about all of its possible consequences.

Consequence-focused instruction group

We are interested in **how you would predict future economic conditions under a certain economic environment**. On the next page, you will be asked to consider a scenario first and then asked to make a series of predictions for future economy. There is no right or wrong answers to the predictions.

Below is our instruction, **please read it carefully.**

Instruction:

We have found that people's predictions are often most accurate when they think thoroughly about the potential consequences of an event, rather than simply based on their feelings or emotions. Therefore, when making the following predictions on the next page, try to **think carefully about what consequences it would have if the event described were to occur**, and base your predictions on the desirability of these consequences.

No specific instruction group

We are interested in **how you would predict future economic conditions under a certain economic environment**.

On the next page, you will be asked to consider a scenario first and then asked to make a series of predictions for future economy. There is no right or wrong answers to the predictions.

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Inflation due to higher demand

First, we would like you to think about the following scenario.

Scenario: inflation rate *unexpected* rises by 10 percentage points due to higher demand

Imagine the inflation rate in US ***unexpected*** rises by 10 percentage points further *over the next 12 months* due to a higher demand for goods and services in the US. That is, an average bundle of goods and products that was expected to cost $1000 in 12 months now will cost $1100.

Inflation due to oil supply chain disruption

First, we would like you to think about the following scenario.

Scenario: inflation rate *unexpected* rises by 10 percentage points due to disruption in oil supply chains

Imagine the inflation rate in US ***unexpected*** rises by 10 percentage points further *over the next 12 months,* mainly driven by **severe disruption in oil supply chains**, due to the escalated Russian-Ukraine war. That is, an average bundle of goods and products that was expected to cost $1000 in 12 months now will cost $1100.

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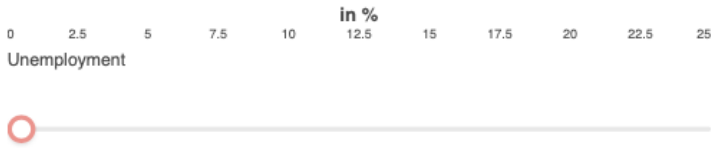
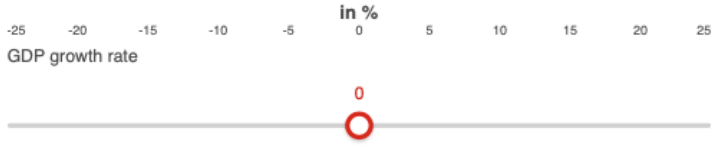
Post-instruction predictions: using consequence-focused instruction group as an example

Q4. Under this scenario and based on careful consideration of all consequences, what do you think the **unemployment rate** in the US would be over the next 12 months from now?



Q5. Under this scenario and based on careful consideration of all consequences, what do you think the **GDP growth rate** in the US would be over the next 12 months from now?



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